

Mediterranean and low-fat diets improve endothelial function in hypercholesterolemic men.

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BACKGROUND: The regulatory function of the endothelium is altered in hypercholesterolemia, and the subsequent endothelial dysfunction plays a central role in the development of atherosclerosis. **OBJECTIVE:** To determine whether endothelial function in hypercholesterolemic patients is affected by replacing a saturated fat-enriched diet with a low-fat, low-saturated fat diet (the U.S. National Cholesterol Education Program stage 1 [NCEP-1] diet) or a diet rich in monounsaturated fat (such as that common in Mediterranean countries). **DESIGN:** Intervention dietary study with a baseline phase and two randomized crossover dietary periods. **SETTING:** Hospital Universitario Reina Sofía, Córdoba, Spain. **PATIENTS:** 22 hypercholesterolemic men. **INTERVENTION:** Patients followed a diet high in saturated fat, then were assigned in a crossover design to the NCEP-1 diet or a Mediterranean diet. Each dietary period lasted 28 days. **MEASUREMENTS:** Plasma P-selectin levels, lipid concentrations, and endothelial function. **RESULTS:** Compared with the saturated fat diet, flow-mediated dilatation increased during the Mediterranean diet but not during the NCEP-1 diet. In addition, levels of plasma cholesterol, low-density lipoprotein cholesterol, apolipoprotein B, and P-selectin decreased during the NCEP-1 and Mediterranean diets. **CONCLUSION:** In hypercholesterolemic men, diets low in fat (especially saturated fat) and diets rich in monounsaturated fats improve endothelial function.